

Model-Based Meta-Analysis of Oral Acetaminophen Pharmacokinetics

Using Published Aggregate Data with the admixr2 R package

Project Documentation — Step-by-Step Methods

Author	Pramod BR
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Project stage	Steps 0-4 complete (setup, review, extraction, dataset, model)
Core tool	admixr2 0.2.0 (nlmixr2 / rxode2 ecosystem)

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1. Project Overview and Objectives

This project builds a **model-based meta-analysis (MBMA)** of the pharmacokinetics (PK) of oral immediate-release acetaminophen (paracetamol) in healthy adults. Instead of collecting new clinical data, it re-uses **summary-level results already published** in the scientific literature — mean concentration-time profiles and PK parameters — and fits a single, unified PK model to them.

By the end of the project the following will be delivered:

- A systematic literature review with explicit, reproducible selection criteria.
- A structured dataset of extracted PK summary data.
- A structural PK model fitted to the aggregate data with the **admixon2** package.
- An assessment of between-study variability and of dose and covariate effects.
- Publication-quality diagnostic and result figures.
- The complete, reproducible workflow published on GitHub.

Why a meta-analysis instead of a new study?

Individual patient data are expensive and often unavailable. A great deal of PK information, however, already sits in published papers as means and standard deviations. A meta-analysis pools that evidence, giving a wider dose range, more subjects, and a more robust picture of variability than any single study could provide.

Why *model-based*, and why aggregate data?

A naive meta-analysis just averages reported numbers. A **model-based** meta-analysis instead fits a mechanistic PK model (absorption, distribution, elimination) directly to the summary statistics. That is exactly what **admixon2** enables: it treats each study's *mean vector* and *covariance* as the observation and matches them against model predictions. This keeps the pharmacological structure intact while using only the aggregate data that publications actually report.

2. How the Project Fits Together

The workflow is a linear pipeline. Each script consumes the output of the previous one and writes a well-defined artifact for the next. This makes the whole analysis reproducible: re-running the scripts in order regenerates every result from the raw extracted data.

Step	Script	Input	Output
0 Setup	00_setup.R	(installed R)	packages loaded, paths, sessionInfo.txt
1 Review	docs (review + log)	literature	included/excluded studies
2 Extract	02_data_extraction.R	raw CSV of PK values	pk_summary_clean.csv
3 Build	03_build_dataset.R	clean summary + profiles	mbma_dataset.csv
4 Fit	04_fit_model.R	mbma_dataset.csv	fitted model + parameter table
5 Covariates	05_covariate_analysis.R	fitted model	dose/covariate effects (next)
6 Plots	06_plots.R	fitted model	publication figures (next)

Table 1. The end-to-end pipeline. Steps 0-4 are built; 5-6 are the next stages.

The software stack

Three R packages do the modelling work, layered on top of one another:

- **rxode2** — the engine that solves the differential equations describing how drug moves through the body (dose → absorption → central compartment → elimination).
- **nlmixr2** — the population non-linear mixed-effects layer built on rxode2; the standard for population PK in R.
- **admixr2** — sits on top and lets those models be fitted to *aggregate* (summary-level) data instead of individual records. This is the heart of the project.

3. Project Folder Structure

A clean, conventional layout keeps raw data, code, and outputs separate — a prerequisite for reproducibility and for publishing on GitHub.

```
MBMA_Acetaminophen_PK/  
  data/  
    raw/          extracted values, digitised profiles (never edited by code)  
    processed/    cleaned, analysis-ready dataset  
  scripts/        00_setup ... 06_plots R scripts  
  models/         saved fitted model objects (.rds)  
  results/        parameter tables, diagnostics, sessionInfo.txt  
  plots/          publication-quality figures  
  docs/           protocol, literature review, this document  
  README.md       objectives + workflow  
  .gitignore       keeps large/regenerable files out of Git  
  MBMA_Acetaminophen_PK.Rproj  RStudio project anchor
```

The **.Rproj** file matters more than it looks: opening it sets the working directory to the project root, so every script finds data/, models/, etc. through the here package without any hard-coded machine-specific path.

4. Step 0 — Environment Setup (00_setup.R)

This script establishes a **reproducible environment** before any data is touched. It must be run as a whole (RStudio → Source), because each line depends on the ones above it.

What it does, in order

- **Installs** any missing packages (rxode2, nlmixr2, admixr2, tidyverse, readxl, here) — safe to re-run.
- **Loads** them, and **stops with a clear message** if any fail, so problems surface immediately.
- **Creates** the project folders if absent and defines a central paths list.
- **Records** the exact package versions to results/sessionInfo.txt.
- **Sets a random seed** so the Monte-Carlo estimator later is reproducible.

```
pkgs <- c("rxode2", "nlmixr2", "admixr2", "tidyverse", "readxl", "here")  
missing <- setdiff(pkgs, rownames(installed.packages()))  
if (length(missing)) install.packages(missing)  
# ... load, stop-if-failed, create folders, write sessionInfo, set.seed(20260711)
```

Note: rxode2 and nlmixr2 compile C code, so the first install needs a compiler toolchain (Rtools on Windows, Xcode command-line tools on macOS). A confirmed run prints: *“Setup complete. All 6 packages loaded. admixr2 0.2.0”*.

5. Step 1 — Systematic Literature Review

The review decides **which studies enter the analysis and why**. The selection rules were fixed in advance (a-priori) so the process is transparent and reproducible.

Eligibility criteria (PICO)

Element	Rule for this project
Population	Healthy adults (≥ 18 y); special groups excluded from the primary set
Intervention	Oral immediate-release (IR) acetaminophen: conventional tablet or capsule
Comparator	None (descriptive PK); IV arms used only to derive absolute bioavailability
Outcome	Reported PK summary (Cmax, Tmax, AUC, $t_{1/2}$, CL/F, V/F, F) and/or a mean conc-time profile

Why formulation and population are restricted

Effervescent, buffered, sustained-release and liquid forms change the *absorption* process — the very thing the model tries to estimate cleanly — so they are excluded. Likewise, disease and frailty alter clearance, so the base dataset is healthy adults only; age, food and sex are kept as covariates to be tested rather than sources of uncontrolled noise.

Result

Eighteen candidate records were catalogued; **9 were included** and 9 excluded (3 of those kept as reference sources rather than data rows). The included set spans 500, 650, 1000 and 2000 mg, single- and multiple-dose, tablet and capsule, both sexes, fasted and fed, and both classic (1977-1983) and modern (2003-2018) studies — the spread needed to identify the model and test a dose effect.

Decision	Count	Basis
Include	9	Healthy-adult IR oral PK with extractable data
Exclude – non-IR formulation	4	Effervescent, buffered, syrup, suspension
Exclude – population/disease	2	Frail elderly, malaria patients
Exclude – confounded absorption	1	Opioid co-administration, no control arm
Exclude – no primary data	3	Reviews/regulatory (kept as references)

Table 2. Screening outcome. Full per-paper rationale is in literature_screening_log.xlsx.

Honesty note. This was a rapid web-based scoping search, not yet a formal indexed PubMed + Embase export. Before publication it should be re-run as a formal database search with recorded hit counts and two-reviewer screening; the log is structured to accept those final numbers.

6. Step 2 — Data Extraction (02_data_extraction.R)

This step turns 9 papers into one tidy table: one row per *study-arm-dose*, one column per standardised quantity. Two rules protect the analysis.

Rule 1 — one unit system

Papers mix units ($\mu\text{g/mL}$ vs mg/L ; mL/min vs L/h). Everything is standardised on entry to **mg/L**, **mg·h/L**, **hours**, **L/h**. Note that 1 mg/L equals 1 $\mu\text{g/mL}$, so concentrations usually need no numeric change, only relabelling; clearance in mL/min is multiplied by 0.06 to get L/h .

Rule 2 — never invent numbers

Each row carries a `data_status` flag: **extracted** (value transcribed and sourced) or **to_extract** (the paper qualifies but the number must still be read off its PDF/figure). Only verified values are filled in; nothing is guessed. Every row also carries a `source_ref`.

What the script checks

- Required columns are present.
- No impossible values (dose ≤ 0 , negative C_{max} , bioavailability outside 0-1).
- Reports how many rows are transcribed vs still pending, then writes `pk_summary_clean.csv`.

7. Step 3 — Build the MBMA Dataset (03_build_dataset.R)

The fitting engine needs the concentration-time *profiles* in aggregate form. This step assembles them into the analysis dataset.

The key idea: aggregate data must carry N and SD

In an MBMA the **study** is the unit of observation. How much a study counts depends on how precisely its mean was measured, so every profile point carries three numbers: the **mean** concentration, its **standard deviation**, and the **sample size N**. The script also computes the standard error of the mean ($SD/\sqrt{N}$) — the precision the model weights by. A 40-subject study rightly outweighs a 6-subject one.

Where the profiles come from

Published figures are digitised with a tool such as **WebPlotDigitizer**: load the figure, calibrate the axes, click along the mean curve to capture (time, concentration) pairs, and read the error bars for the SD. These are pasted into `data/raw/pk_profiles.csv`. The script drops the example rows, validates the data, prints dose coverage, and writes `mbma_dataset.csv`.

8. Step 4 — Fit the Structural Model (04_fit_model.R)

This is where the pharmacology meets the data. A structural PK model is defined and fitted to the aggregate dataset with `admixr2`.

The structural model

The starting point is a **one-compartment model with first-order absorption** — the standard first model for an oral drug. Drug enters a depot (the gut), is absorbed into the central compartment at rate k_a , and is eliminated with clearance CL from volume V .

```
d/dt(depot) = -ka * depot
d/dt(central) = ka * depot - (CL/V) * central
cp = central / V      # predicted plasma concentration
```

Two deliberate modelling choices:

- **Apparent parameters (CL/F, V/F).** With oral-only data you cannot separate clearance from bioavailability F , so the identifiable quantities are CL/F and V/F . This is a fact of the data, not a shortcut.
- **Mu-referencing.** Each parameter is written as $\exp(\theta + \eta)$ on the log scale. This keeps values positive and is the exact form `admixr2` needs to compute analytical gradients efficiently.

How admixr2 receives the data

For each study arm the script builds the five objects admixr2 expects and packs them into a named studies list:

Object	Meaning
E	mean concentration at each observation time
V	variance (SD ²) of those means — here diagonal, since papers rarely report cross-time covariance
n	sample size of the study arm
times	the observation time points
ev	the dosing event (single oral dose = dose_mg)

```
fit <- nlmixr2(oral_1cmt, admData(), est = "admc",
  control = admControl(studies = studies, n_sim = 3000L,
    cov_n_sim = 6000L, maxeval = 300L, seed = 1L))
```

Choosing an estimator

admixr2 offers four estimators. They trade speed against exactness:

Estimator	est =	When to use
First-Order	adfo	Fast screening and initial estimates
Monte Carlo	admc	The workhorse for standard PK models (used here)
Gauss-Hermite	adgh	Noise-free alternative for models with few random effects
Iterative Reweighting MC	adirmc	Complex ODE systems, expensive solves, poor starting values

Table 3. The four admixr2 estimators.

Outputs

The script prints the fit, back-transforms the log-scale estimates into readable **ka**, **CL/F** and **V/F**, reports the objective function (-2 log-likelihood) and AIC, then saves the fitted object to `models/` and the parameter table to `results/`. Starting values are literature-typical for acetaminophen: $ka \approx 1.5/h$, $CL/F \approx 21 \text{ L/h}$ ($\approx 350 \text{ mL/min}$, matching Rawlins 1977), $V/F \approx 60 \text{ L}$.

Diagonal-variance caveat. Because publications rarely give the cross-time covariance, V is used as a diagonal variance vector (SD² per time). This is a standard, slightly conservative approximation and should be stated in the write-up.

9. What Comes Next

With the base model in place, the remaining stages are:

- **Two-compartment alternative** — acetaminophen can show a distribution phase; fit a 2-compartment model and compare by AIC.
- **Step 5 — Covariate analysis** (05_covariate_analysis.R): test dose, formulation (tablet vs capsule), food, age and sex; quantify how much between-study variability each explains.
- **Step 6 — Plots** (06_plots.R): observed-vs-predicted profiles, goodness-of-fit, and covariate effect figures for publication.
- **Publish** the reproducible workflow on GitHub.

10. Glossary

Term	Meaning
PK	Pharmacokinetics — what the body does to a drug over time
MBMA	Model-based meta-analysis — fitting a mechanistic model to pooled summary data
Aggregate data	Study-level means, SDs and N (no individual records)
C _{max} / T _{max}	Peak concentration and the time at which it occurs
AUC	Area under the concentration-time curve — total exposure
t _{1/2}	Elimination half-life
k _a	First-order absorption rate constant
CL/F, V/F	Apparent clearance and volume (clearance/volume divided by bioavailability F)
F	Bioavailability — fraction of an oral dose reaching the circulation
Mu-referencing	Writing a parameter as exp(theta + eta) so fixed and random effects add on the log scale
IIV	Between-subject (inter-individual) variability
AIC	Akaike Information Criterion — compares models, penalising complexity

Companion files in this project: *README.md*, *docs/01_systematic_literature_review.md*, *docs/literature_screening_log.xlsx*, and the *scripts/* folder (00-04). Full per-paper sources and DOIs are in the screening log.